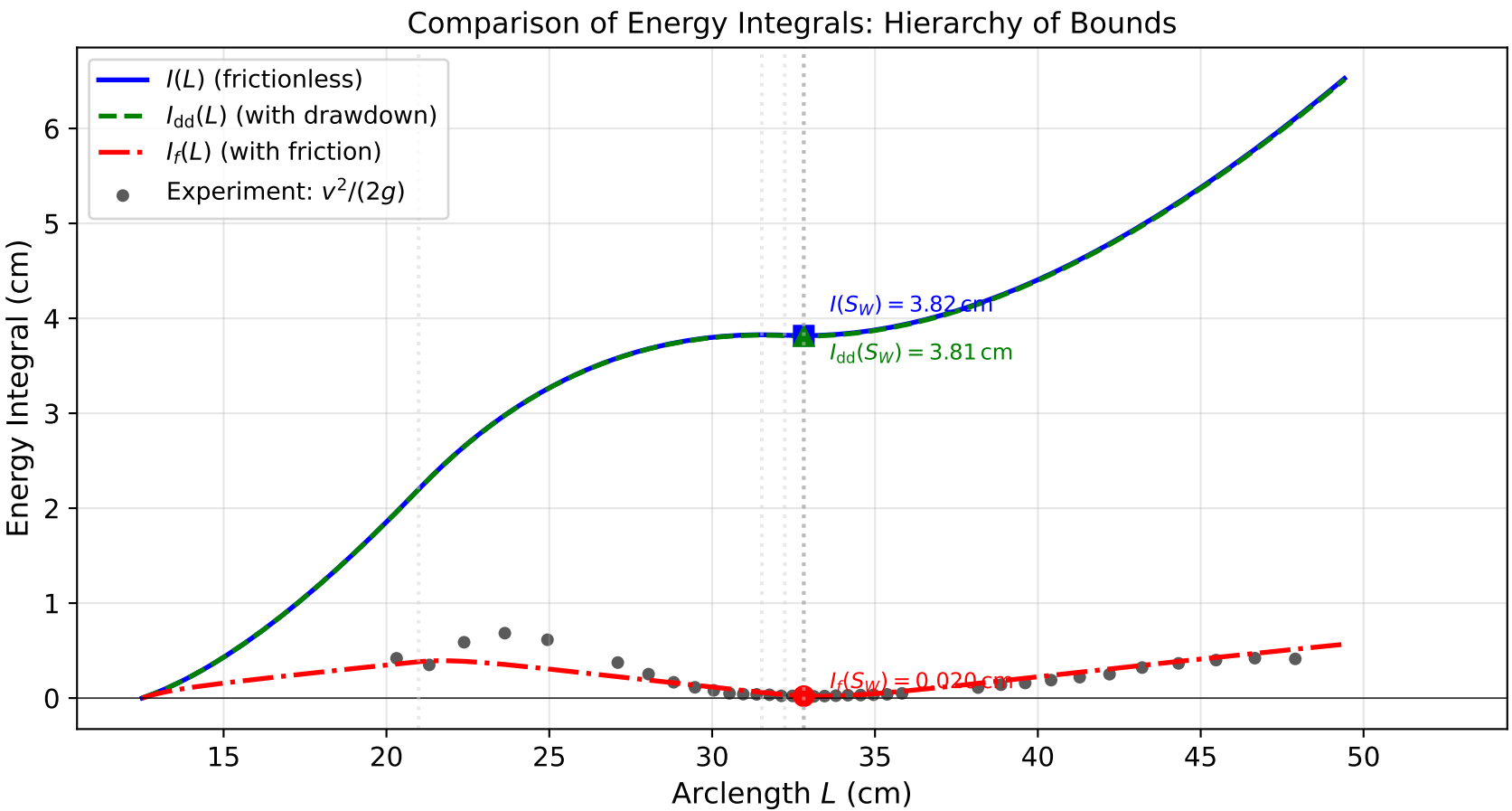
Comparison of Energy Integrals: Hierarchy of Bounds
I(L) (frictionless)
I_dd(L) (with drawdown)
I_f(L) (with friction)
Experiment: v²/(2g)
I(S_W) = 3.82 cm
I_dd(S_W) = 3.81 cm
I_f(S_W) = 0.020 cm
Energy Integral (cm)
Arclength L (cm)
0
1
2
3
4
5
6
15
20
25
30
35
40
45
50